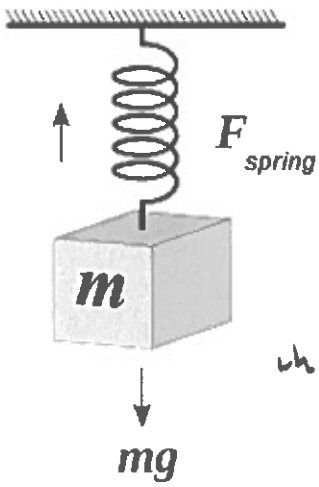
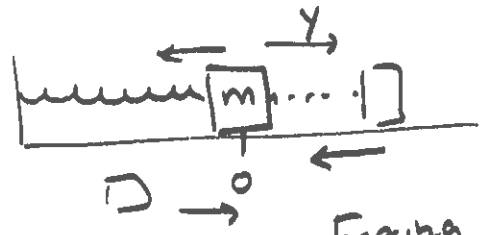
L11 The Mass-Spring Oscillator

Section 4.1

A damped mass-spring oscillator consists of a mass m attached to a spring fixed at one end.





$F = ma$

$m \cdot a = -ky$ ← F_{spring} Hooke's Law (springs)

$my'' = -ky$

include ~~damp~~ damping:

$my'' = -ky - \underbrace{by'}_{\text{damping}} \Rightarrow my'' + by' + ky = 0$

The differential equation that governs the motion of this oscillator can be derived from Newton's second law ($F = ma$) and is given by

$$my'' + by' + ky = \underbrace{F_{ext}} \quad (1)$$

where y is displacement from equilibrium

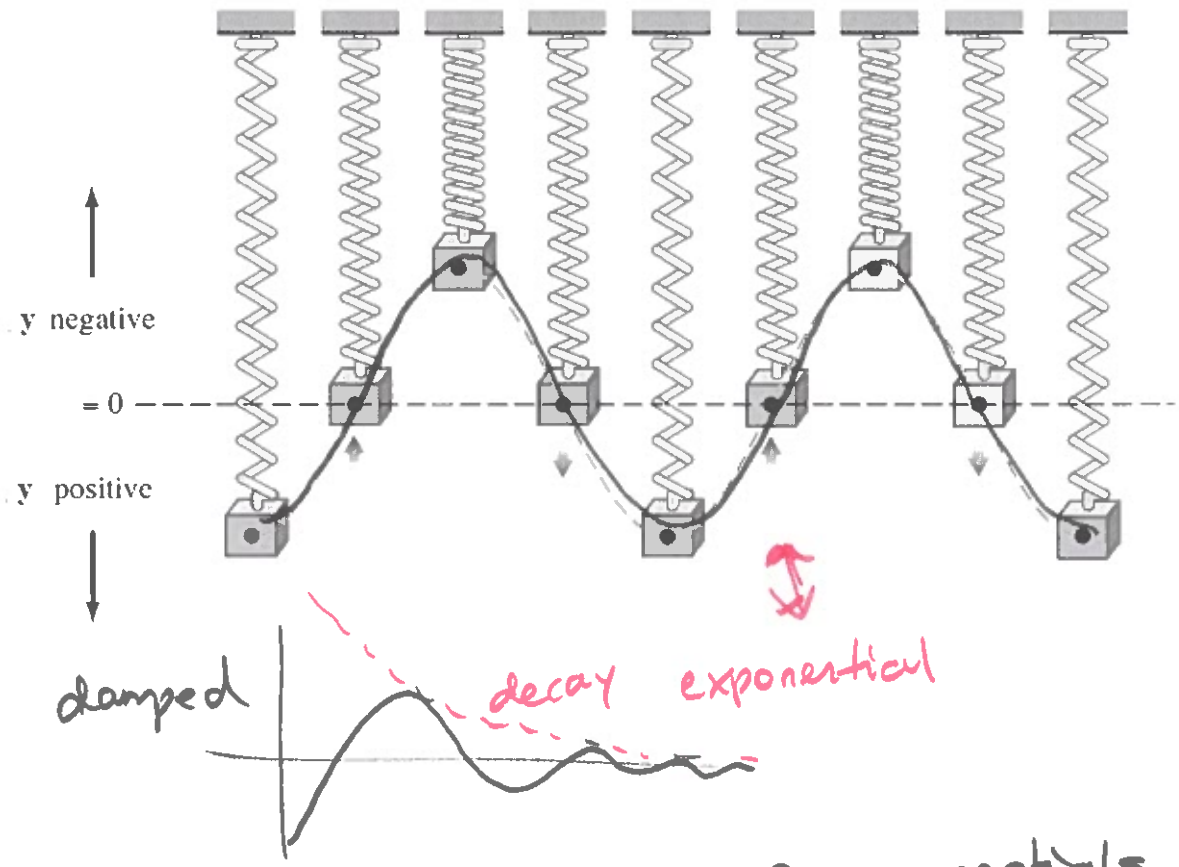
m is the mass of the spring

k is the spring constant, also known as stiffness, from Hooke's Law

b is a damping coefficient (friction, opposing the motion of the spring)

F_{ext} is the sum of all externally applied forces.

Spring motion



Solutions to be combinations of exponentials and trigonometric functions.

no damping

Spring motion when $b = 0$, $F_{ext} = 0$.

Consider the case where $b = 0$ and $F_{ext} = 0$. Equation (1) simplifies to

divide by m

$$my'' + ky = 0 \longrightarrow y'' + \frac{k}{m}y = 0$$

In this case, we expect the displacement to oscillate perpetually with no drop-off, resembling a sine or cosine graph. The oscillations should speed up with stiffer springs but slow down for heavier masses.

ex. Verify that if $b = 0$ and $F_{ext}(t) = 0$; the equation has a solution of the form $y(t) = \cos(\omega t)$ and that the angular frequency ω increases with k and decreases with m.

frequency

$y = \cos(\omega t)$

$y' = -\omega \sin(\omega t)$ $\qquad$ $y'' + \frac{k}{m} y = 0$

$y'' = -\omega^2 \cos(\omega t)$

$\Rightarrow (-\omega^2 \cos(\omega t)) + \frac{k}{m}(\cos(\omega t)) = 0$

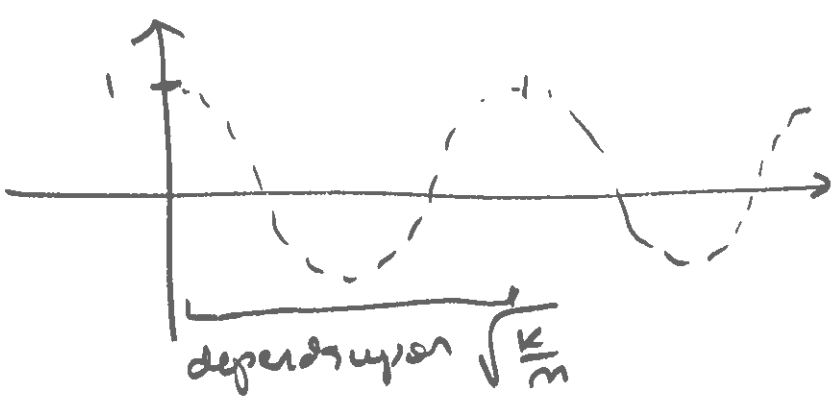
$\Rightarrow \left(\frac{k}{m} - \omega^2\right)\cos(\omega t) = 0$

$\Rightarrow \frac{k}{m} - \omega^2 = 0 \Rightarrow \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$

we have that $y = \cos\left(\sqrt{\frac{k}{m}}\, t\right)$ solves the equation.

$\sqrt{\frac{k}{m}}$ decreases if m gets larger (more mass)

increases if k gets larger (stiffer)



we also have $y = \sin(\omega t)$ as a solution.

all of the solutions are of the form

$C_1 \cos(\omega t) + C_2 \sin(\omega t)$

$b > 0$

Spring motion when $b \neq 0$, $F_{ext} = 0$.

When damping is present, the oscillations eventually die out, so the solution will take on a different shape. Systems with low damping, such as tuning forks, continue to vibrate for some time, while those with high damping, like car suspension systems, use friction to help suppress or eliminate oscillations. Sometimes this distinction is described as underdamped (low damping) versus overdamped (high damping) system.

$$my'' + by' + ky = 0$$

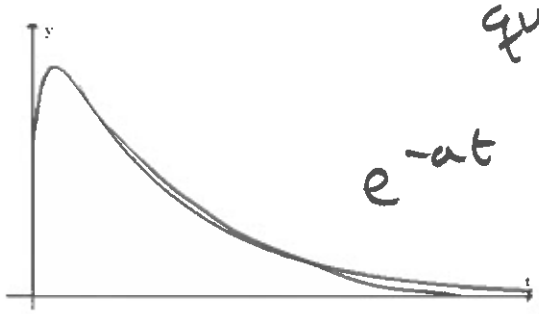
quadratic: $mx^2 + bx + k = 0$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

$b^2 - 4mk$

discriminant

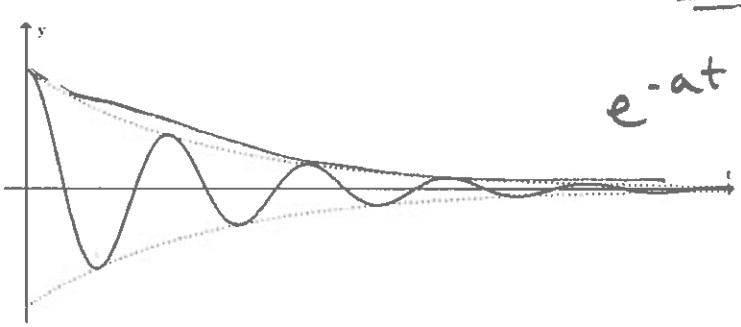
Overdamped System.



e^{-at}

Overdamping occurs when the frictional force exceeds the spring force, i.e. $b^2 - 4mk > 0$. There are no oscillations.

In the **Underdamped** system friction is present, but is not large enough to supress oscillations. Here, $b^2 - 4mk < 0$.



$e^{-at}(\cos(\omega t) + \sin(\omega t))$

ex. Verify that the function $y(t) = e^{-3t}$ is a solution to the mass-spring equation if $F_{ext} = 0$, $m = 1$, $b = 5$, and $k = 6$:

$$my'' + by' + ky = 0 \Rightarrow y'' + 5y' + 6y = 0$$

$$b^2 - 4mk = 5^2 - 4(1)(6) = 25 - 24 = 1 > 0$$

overdamped.

Show $y = e^{-3t}$ is a solution:

$$y' = -3e^{-3t}$$

$$y'' = 9e^{-3t}$$

$$y'' + 5y' + 6y = (9e^{-3t}) + 5(-3e^{-3t}) + 6(e^{-3t})$$

$$= 9e^{-3t} - 15e^{-3t} + 6e^{-3t} = 0 \checkmark$$

Quadratic: $x^2 + 5x + 6 = 0 \Rightarrow (x+2)(x+3) = 0$

$$\Rightarrow x = -2, -3$$

Try $y = e^{-2t}$ is also a solution!

$$y' = -2e^{-2t}$$

$$y'' = 4e^{-2t}$$

$$4e^{-2t} + 5(-2e^{-2t}) + 6e^{-2t} = 4e^{-2t} - 10e^{-2t} + 6e^{-2t}$$

$$= 0 \checkmark$$

General solution $c_1 e^{-2t} + c_2 e^{-3t}$

ex. Verify that damped sinusoid $y(t) = e^{-3t}\cos(4t)$ is a solution to the mass- spring equation if $F_{ext} = 0$, $m = 1$, $b = 6$, and $k = 25$.

$$my'' + by' + ky = 0 \Rightarrow y'' + 6y' + 25y = 0$$

$$b^2 - 4mk = 6^2 - 4(1)(25) = 36 - 100 = -64 < 0$$

<u>underdamped</u>.

$$y = e^{-3t}\cos(4t)$$

$$y' = -3e^{-3t}\cos(4t) + e^{-3t}(-4\sin(4t))$$

$$y'' = [9e^{-3t}\cos(4t) - 3e^{-3t}(-4\sin(4t))]$$

negative

$$+ [-3e^{-3t}(-4\sin(4t)) + e^{-3t}(16\cos(4t))]$$

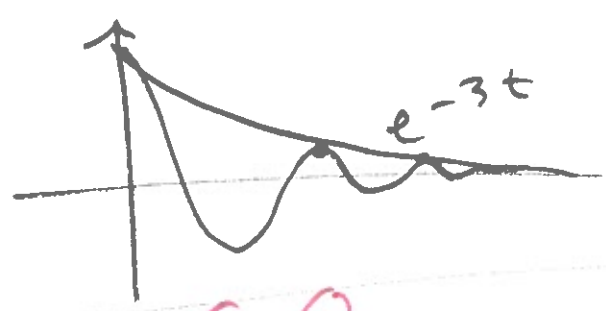
$$y'' + 6y' + 25y = [9e^{-3t}\cos(4t) + 24e^{-3t}\sin(4t) - 16e^{-3t}\cos(4t)]$$

$$+ 6(-3e^{-3t}\cos(4t) - 4e^{-3t}\sin(4t))$$

$$+ 25(e^{-3t}\cos(4t))$$

$$= e^{-3t}\cos(4t)(9 - 16 - 18 + 25) + e^{-3t}\sin(4t)(24 - 24)$$

$$= 0 \checkmark$$



$$x^2 + 6x + 25 = 0$$

$$x = \frac{-6 \pm \sqrt{36 - 4(1)(25)}}{2}$$

$$= \frac{-6 \pm \sqrt{-64}}{2} = -3 \pm \frac{8i}{2} = -3 \pm 4i$$

External Forces

Typically external forces are sinusoidal with fixed amplitude, and the resulting solution curves will eventually synchronize with these forces and oscillate at the same frequency, though the initial behavior can be erratic. Examples of such systems are sound system speakers, electronic amplier circuits, and ocean tides (affected by the moon.) It is important to observe that some systems are incredibly sensitive to the resonant frequency: this is the explanation for how perfectly tuned notes can shatter crystal and wind-driven vibrations can destroy bridges. This is a major point to consider when engineers are designing systems, whether you want to avoid such resonant responses or not (as with radio signals).

ex. Find the synchronous response of the mass-spring oscillator with $F_{ext} = 5\sin(\omega t)$, $m = 1$, $b = 2$, and $k = 4$, and $\omega = 3$.

$$my'' + by' + ky = F_{ext} \Rightarrow y'' + 2y' + 4y = 5\sin(3t)$$

expect solution $y = C_1\cos(3t) + C_2\sin(3t)$

$$y' = -3C_1\sin(3t) + 3C_2\cos(3t)$$

$$y'' = -9C_1\cos(3t) - 9C_2\sin(3t)$$

$$\Rightarrow \overset{y''}{(-9C_1\cos(3t) - 9C_2\sin(3t))} + 2\overset{y'}{(-3C_1\sin(3t) + 3C_2\cos(3t))}$$
$$+ 4\underset{y}{(C_1\cos(3t) + C_2\sin(3t))} = \underset{F_{ext}}{5\sin(3t)}$$

$$\Rightarrow \cos(3t)(-9C_1 + 6C_2 + 4C_1) + \sin(3t)(-9C_2 - 6C_1 + 4C_2)$$
$$= 5\sin(3t)$$

$$\Rightarrow (6C_2 - 5C_1)\cos(3t) + (-6C_1 - 5C_2)\sin(3t) =$$
$$0\cdot\cos(3t) + 5\sin(3t)$$

$$\Rightarrow \begin{cases} 6C_2 - 5C_1 = 0 \\ -6C_1 - 5C_2 = 5 \end{cases} \longrightarrow 6C_2 = 5C_1 \Rightarrow C_2 = \frac{5}{6}C_1$$

$$-6C_1 - 5\left(\frac{5}{6}C_1\right) = 5 \Rightarrow -6C_1 - \frac{25}{6}C_1 = 5$$

$$\frac{-36-25}{6}C_1 = \frac{-61}{6}C_1 = 5$$

$$\boxed{C_1 = -\frac{30}{61}}$$

$$\Rightarrow \boxed{C_2 = \frac{5}{6}C_1 = \frac{5}{6}\cdot\frac{-30}{61} = \frac{-25}{61}}$$

So $\boxed{y = -\frac{30}{61}\cos(3t) - \frac{25}{61}\sin(3t)}$

ex.

Consider the undamped case where a periodic forcing function is present that has an input frequency ω close to the spring's natural angular frequency ω_0:

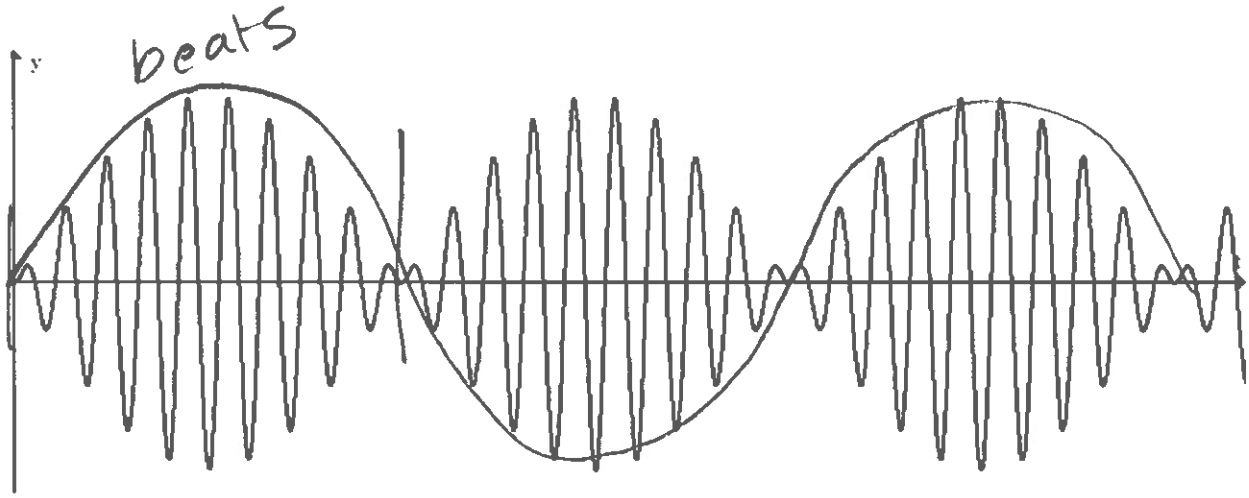
$F_{ext} = 0$

$my'' + ky = 0 \rightarrow \omega_0$

$my'' + ky = F_{ext}$

ω

We can see this below:



When ω is close to the spring's natural frequency ω_0, **beats** occur. Piano tuners use beats to tune pianos[1]. When a tuning fork is struck at the same time as a piano string at almost the same frequency, the resulting beats are audible. The tuner adjusts the tension in the piano wire to slow down the beat frequency and eliminate it.

[1] https://youtu.be/Lb-GD0g1ADA

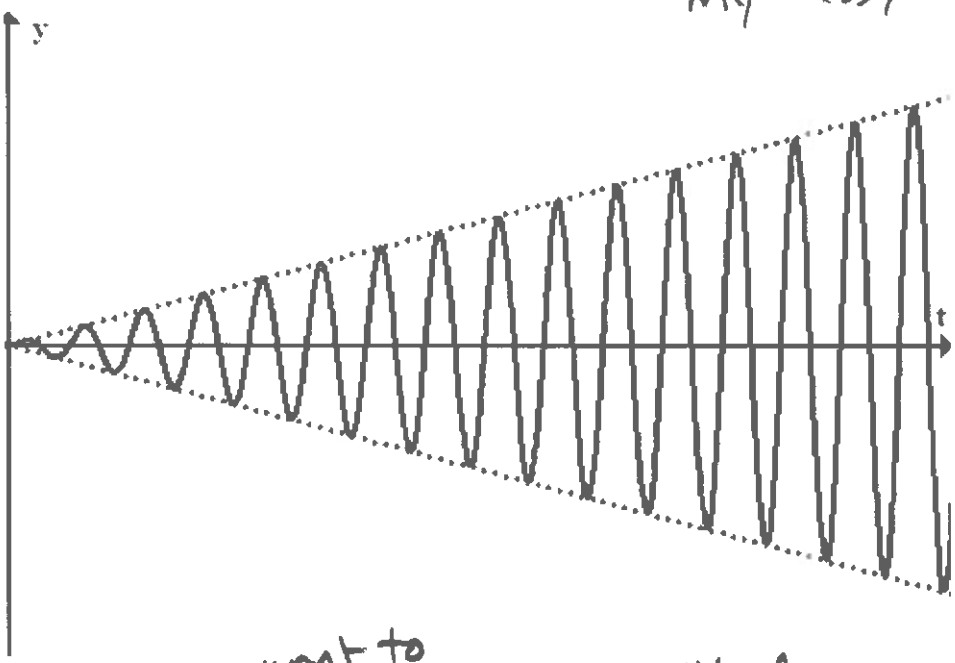
Resonance occurs where a periodic forcing function is present that has a driving frequency ω **exactly equal** to the spring's natural frequency ω_0.

$my'' + by' + ky = 0 \leftarrow \omega_0$

$my'' + by' + ky = F_{ext}$ //

$\uparrow$

ω



useful when want to increase amplitude

Some applications of resonance:

- Seismographs take advantage of resonance to detect tremors in the earth.
- Nuclear Magnetic Resonance in medical imaging.
- Fluorescent lighting
- AM radio: tune in to a carrier frequency by adjusting a variable capacitor which changes the frequency of a simple RLC circuit so as to agree with ω_0.
- Microwave ovens
- Feedback from electric guitars
- Clocks keeping time with quartz crystals
- Foucault Pendulums
- Opera singers breaking glass[2]

When resonance is bad:

Mechanical systems and structures like buildings or bridges could experience violent swaying and catastrophic failure. Avoiding such disasters is a major concern in building and bridge construction.

[2] https://youtu.be/TbuqKKvcziA

Harmonic oscillators occur in a number of different engineering areas. The mathematical models are identical but the variables and constants represent different physical phenomena. Examples include the following:

Rotational mechanics:

$I\frac{d^2\theta}{dt^2} + \Gamma\frac{d\theta}{dt} + \mu\theta = \tau$ where

$I =$moment of interta

$\mu =$torsion constant

$\Gamma =$ rotational friction

$\tau =$ drive torque

$\theta =$rotational angle

Series RLC circuit:

$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = e$ where

$L =$inductance

$q =$charge

$C =$capacitance

$R =$resistance

$e =$voltage